Essential oils/ Volatile oils

By Million

Brainstorming

- Volatile guy
- Words can't be unsaid (The Daily Stoic).pdf



October 5th

WORDS CAN'T BE UNSAID

"Better to trip with the feet than with the tongue."

—ZENO, QUOTED IN DIOGENES LAERTIUS, *LIVES OF THE EMINENT PHILOSOPHERS*, 7.1.26

You can always get up after you fall, but remember, what has been said can never be unsaid. Especially cruel and hurtful things.

Outline

- Introduction
- Physical properties
- Composition
- Methods of preparation
- Uses
- Distribution and occurrence

Definition

- The **odorous, volatile** principle of plant and animal sources are known as volatile oils.
- As they evaporate when exposed to air at ordinary T°, they are called **ethereal oil** or **aetherolea**.
- Are also called essential oils because they **produce essence**.
- E.g. Clove, Eucalyptus, Lavender, Lemon, Orange

Introduction

- They differ entirely in **both chemical and physical** properties from **fixed oils.**
- With the exception of oils such **as oil of bitter almonds,** which are produced by the **hydrolysis of glycosides,** these oils are contained largely as such in the plant.
- They are **secreted in oil cells, in secretion ducts** or **cavities or in glandular hairs.**

Cont'd....

- They are frequently associated with other substances such as gums and resins.



Difference between FO and VO

VOLATILE OILS

- Volatilize at room T°
- Obtained by distillation
- Leave no spot after evaporation
- Can not be saponified
- Terpenes/aromatic compounds
- High refractive index
- Leaves, roots and barks are major sources

FIXED OILS

- Don't get volatiles at room T°
- Obtained by extraction
- after evaporation, leave spot
- Can be saponified
- Esters of fatty acid and glycerol
- Low refractive index
- Seeds

Properties

- Soluble in alcohol, ether, and fixed oils, but insoluble in water.
- Generally liquid
- **Colorless at room T°** (Chamomile- violet colour)
- Have a characteristic odor,
- Have a density less than water *(exception cinnamon, clove…)*
- High refractive index
- Most of them are optically active.



Why plants produce essential oils?



1. **Attraction of pollinators**
2. **Protection**

- Compete for survival
- To prevent water loss
- Treatment for disease (in combination with others for wound)
- Protection from pests and predators,

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የቼሮቂ ጎሳዎች በደቡብ ምሥራቅ አሜሪካ የነበሩ ጥንታውያን የአሜሪካ ሕዝቦች ናቸው። የቼሮቂ ወጣቶች ለዐቅመ አዳም መድረሳቸውን የሚገ ልጡበት አንድ ከባድ ፈተና አላቸው። ወጣቱን አባቱ ይዞት ሊመሻሽ ሲል ወደ ጫካ ይሄዳል። ዓይኑን በጨርቅ ይታሠራል። ምንም ነገር ለማየት አይችልም። ያንን ጨርቅም ከዓይኑ ላይ ለማንሣት አይፈቀ ድለትም። አንድ ቦታ ላይ ይቀመጣል። ከተቀመጠበት ቦታም መንቀ ሳቀስ አይፈቀድለትም።

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የሚመጣብህ?›› አለ አባቱ እየሳቀ። ልጁም ሳቀ።

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Neem

Azadirachta indica

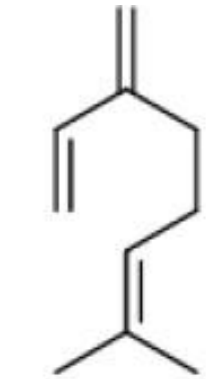


Parts of plants

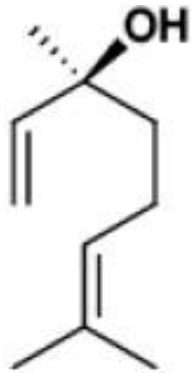
Plant parts	Plants
Flower	lavender, rose, chamomile
Roots	Ginger, plai, turmeric, valerian, vetiver, spikenard, angelica
Stems /barks	Cinnamon Cassia, sassafras, katrafay
Leaves	Eucalyptus, Mentha, cinnamon, common sage, thyme
Wood	Amyris, cedarwood, camphor, rosewood, sandalwood,
Fruits	Xanthoxylum, nutmeg, black peppe
Seeds	Almond, anise, cardamom, caraway, carrot celery, coriander, cumin, nutmeg, parsley, fennel
Peels	Orange, lemon, mandarin
Resin	Frankincense, myrrh
Berries	All spice, juniper
Buds	Clove

Composition

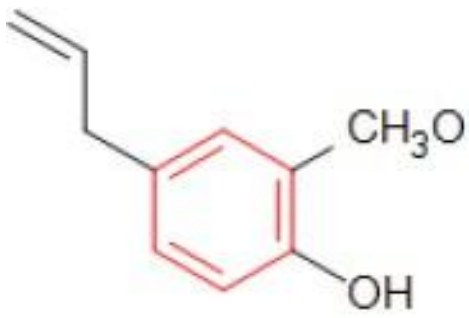
- In general, the constituents in essential oils are
  - **Terpenes** (monoterpenes and sesquerpenes)
  - **Terpenoids** (isoprenoids).
  - **Aromatic compounds**



Myrcene



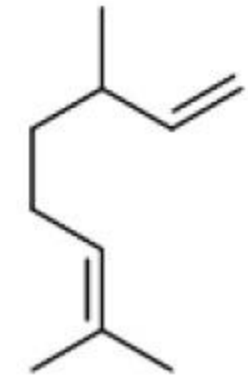
Linalool



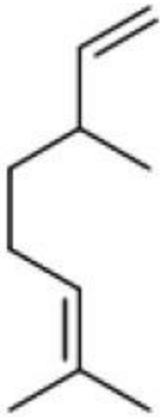
Eugenol

Terpenes and terpenoids

- These hydrocarbons may be **acyclic, alicyclic** (monocyclic, bicyclic, or tricyclic), or **aromatic**.
- Essential oils consist of mainly **monoterpenes** (C10) and **sesquiterpenes** (C15).
- These compounds are the combination of C, H, and O.



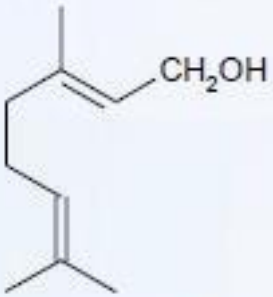
cis-Ocimene



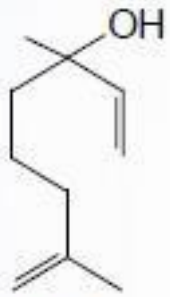
trans-Ocimene

Monoterpene

acyclic

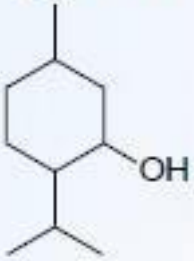


geraniol

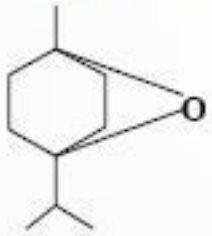


linalool

monocyclic



menthol

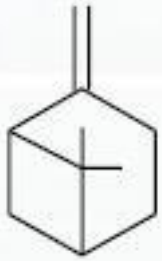


Cyneol

bicyclic

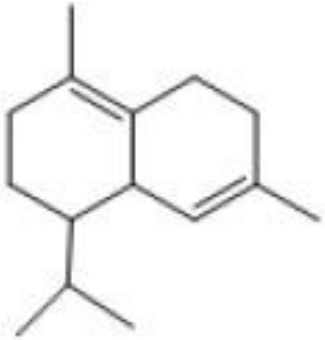


α-pinene

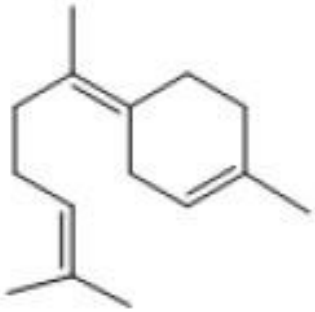


β-pinene

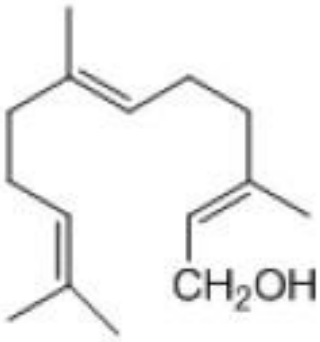
Sesquiterpene



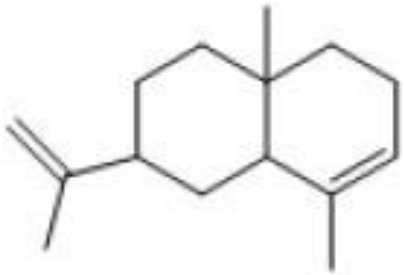
cadinene



bisabolene



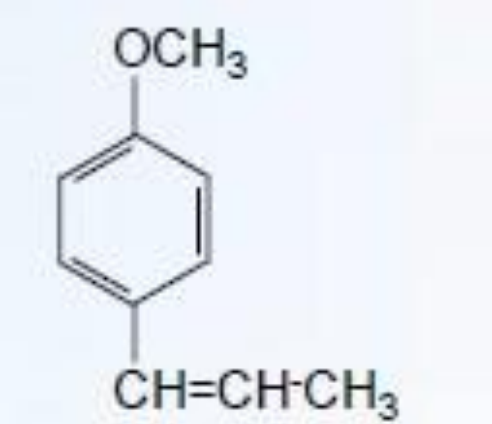
farnesol



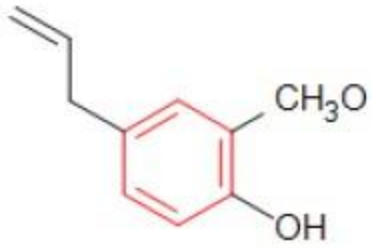
selinene

Phenylpropanoid derivatives

- far less common than terpenoids



Anethole



Eugenol

Aromatic compound extracted from clove essential oils

Functional group

Functional group	Examples
Phenols	***thymol, eugenol***, *carvacrol, chavicol,*
Alcohols	**Geraniol, menthol, Lavundol**
Aldehydes	***citral***, *myrtenal, cuminaldehyde, citronellal,* ***cinnamaldehyde,*** **benzaldehyde**
Ketones	*carvone,* ***menthone****, pulegone, fenchone, camphor, thujone,* verbenone

Cont'd…

Functional group	Examples
Esters:	*bomyl acetate, linalyl acetate, citronellyl acetate, geranyl* acetate,
Oxides	*:* ***1,8-cineole****, bisabolone oxide, linalool oxide, sclareol* oxide
Lactones	*bergaptene, nepetalactone, psoralen, aesculatine, citroptene*
Ethers	*1,8-cineole, anethole, elemicin, myristicin*

Methods of Extraction

- Distillation
- Expression
- Solvent extraction
- Enfluerage

Distillation

Types	Principles
Hydrodistillation	❖ Applied to plant material that is dried and is not destroyed by boiling ❖ Turpentine oil
Water and steam distillation	❖ Fresh or dried plant that can't be destroyed by boiling ❖ Covered with the layer of water ❖ Steam passes through macerated mixture ❖ e.g. Cinnamon or clove
Direct steam distillation	❖ Steam is passed through the fresh herb and carries the oil droplets ❖ Some components are hydrolyzed

Expression

- Expression, or cold-press extraction, is typically used for isolating oils from citrus peels.

Solvent extraction

- Conventional solvent extraction has been implemented for fragile or delicate flower materials,
  - which are not tolerant to the heat of steam distillation.
- Different solvents can be used for extraction
  - acetone, hexane, petroleum ether, methanol, or ethanol
- Solvent is mixed with the plant material and then heated to extract the essential oil, followed by filtration.

- Subsequently, the filtrate is concentrated by solvent evaporation.
- The concentrate is resin (resinoid)
- From the concentrate, it is then mixed with pure alcohol to extract the oil and distilled at low temperatures.
- The alcohol absorbs the fragrance and when the alcohol is evaporated, the aromatic absolute oil is remained.

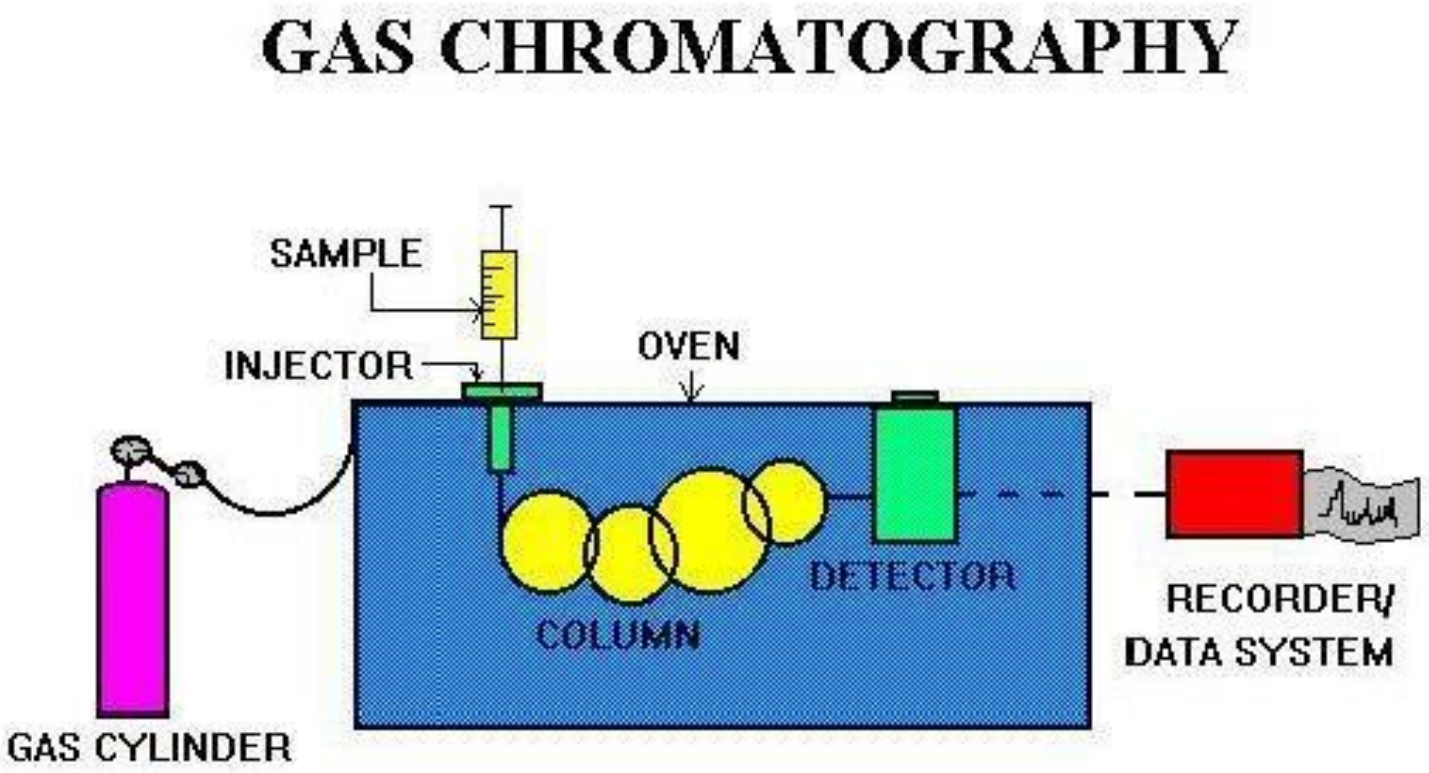
Enfleurage (Extraction of perfume)

- Used for plants decomposed by the action of heat… Present in extremely small amount/ removal of oil is not commercially feasible in other ways.
- Glass plates are covered with thin layer of fixed oil or fat (Tallow & Lard)
- Fresh flowers are spread on them.
- VO gradually passes into the fat
- Exhausted flowers are removed and replaced by fresh supplies.

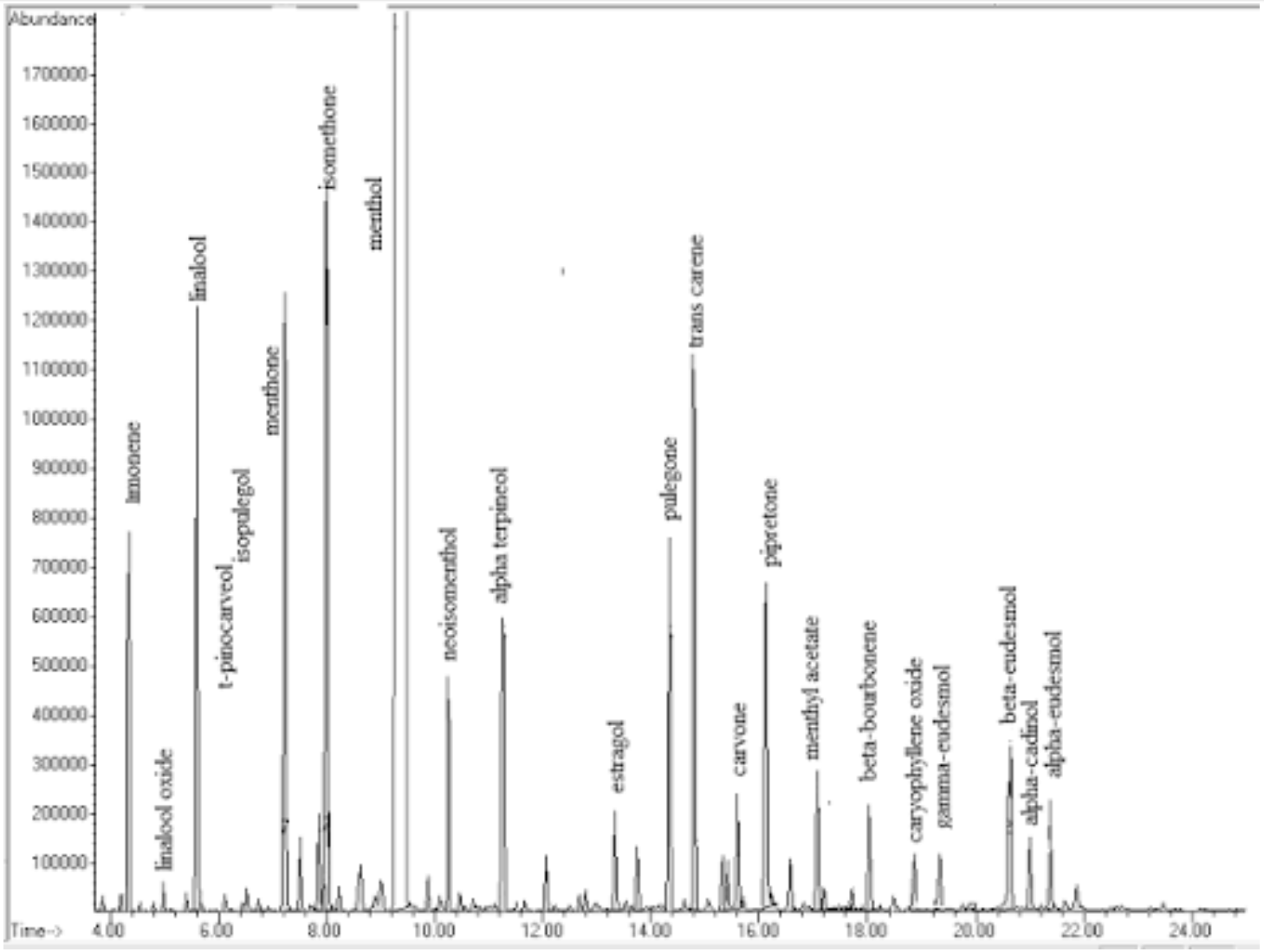
Cont'd...

- Process continues until the fat is saturated.
- The product is known as pomade.
- Pomade:- sold directly or treat with alcohol
- Removal of alcohol and collection of the absolute EO.

GC-MS


GAS CHROMATOGRAPHY
SAMPLE
INJECTOR
OVEN
DETECTOR
COLUMN
RECORDER/
DATA SYSTEM
GAS CYLINDER

Typical GC-MS chromatogram of essential oil from Mentha arvensis



Uses

- Aromatherapy
- Perfumery
- Food additive

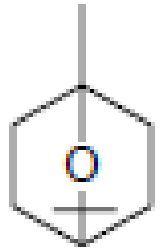
Oil of Rose

- Obtained from the fresh flowers of *Rosa damascene*
- *Chief producing countries are Bulgariam Turkey and Morocco*
- *… A rose is a rose is a rose, unless, of course, it is a Bulgarian rose.*
- *https://youtu.be/26pEG7Ghgpg*
- *>300 compounds are present…* **Citronellol 20–34% Geraniol 15–22%, Nonadecane 8–15%** and others….



Eucalyptus

- Oil of eucalytus is distilled from the fresh leaves of various species of *Eucalyptus (Myrtaceae)*.
- Colorless/pale yellow liquid
- **70% cineole,** 1,8-Cineole and O-cresol.
- Used for alleviating the symptoms of **nasopharyngeal infections, for treating coughs and as a decongestant.**



Cineole

Chemical Composition and Biological Activities of *Eucalyptus globulus* Essential Oil

Natália Čmiková [1,*], Lucia Galovičová [1], Marianna Schwarzová [1], Milena D Vukic [2], Nenad L Vukovic [2], Przemysław Łukasz Kowalczewski [3], Ladislav Bakay [4], Maciej Ireneusz Kluz [5], Czeslaw Puchalski [5], Miroslava Kačániová [1,5,*]

Chemical Composition of *Eucalyptus globulus* Essential Oil.

No	RI [a]	Compound [b]	%
1	909	isobutyl isobutyrate	0.1
2	926	a-thujene	0.4
3	938	a-pinene	7.3
4	948	camphene	0.8
5	977	sabinene	1.0
6	980	b-pinene	3.0
7	992	b-myrcene	1.7
8	1004	a-phellandrene	1.0
9	1009	d-3-carene	0.1
10	1016	a-terpinene	1.0
11	1023	*p*-cimene	7.7
12	1028	a-limonene	6.9
13	1033	1,8-cineole	63.1
14	1047	(*E*)-b-ocimene	0.2
15	1060	g-terpinene	3.6
16	1088	a-terpinolene	0.6
17	1140	trans-pinocarveol	0.1
18	1148	camphor	0.1
19	1151	menthone	0.2
20	1160	pinocarvone	0.1
21	1178	4-terpinenol	0.2
22	1189	a-terpineol	0.1
26	1443	aromadendrene	0.4
27	1498	ledene	tr [c]
	Total		99.7

Table 2.

The volatiles presented in percentage for each class of compounds.

Class of Compounds	%
monoterpenes	99.2
monoterpene hydrocarbons	35.3
oxygenated monoterpenes	63.9
monoterpene epoxide	63.1
monoterpene alcohols	0.4
monoterpene ketones	0.4
sesquiterpenes	0.4
sesquiterpene hydrocarbons	0.4
oxygenated sesquiterpenes	Tr
sesquiterpene alcohols	Tr
non-terpenic	0.1
ester	0.1
Total	99.7

Ocimum

- Obtained from the genus **Ocimum** (family Lamiaceae)
- Main constituents of the oil were found to be **Linalool (28.52%)** and 1-octen-3-yl-npropionate (20.82%) followed by 3,7,11-trimethyl-(E,E)-2,6,10-dodecatrienal (12.14%)



Chemical composition and antimicrobial activity of leave extract of Ocimum lamiifolium (Damakese) as a treatment for urinary tract infection

Nigus Aweke Sahalie, Lijalem Hadush Abrha & **Leta Deressa Tolesa** | Oleg Rakitin (Reviewing editor)

Table 1. Chemical components of *Ocimum lamiifolium leaf* essential oil

Compound name	RT (min)	Mw (g/mol)	Characteristic ions (m/z)	Relative content (area %)
3-Octanol	6.40	130	129, 112, 101, 83, 59, 41	2.66
1-Octen-3-ylpropionate	7.64	184	155, 128, 81, 57, 41	20.82
Linalol	9.03	154	153, 136, 121, 93, 71, 41	28.52
3,7,11-Trimethyl-(E,E)-2,6,10-dodecatrienal	11.07	220	220, 107, 84, 41	12.14
2-Methylphenyl p-methoxybenzoate	11.30	242	242, 152, 135, 77	2.58
7-Methylene bicyclo[3,3,1]nonan-3-ol	12.40	152	152, 134, 119, 92, 79, 41	7.03
2,4,5,6,7,7-Hexahydro-4,7-methano-1H-indene	13.63	134	134, 119, 91,77	6.75
2-Methoxy-4-(1-propenyl)-phenol	15.77	164	164, 121, 91, 134, 105, 91	4.33
δ-Cadinene	19.80	204	204, 161, 134, 105, 91	5.01
α-Terpineol	20.44	154	154, 136, 121, 94, 79, 55	3.72

Download CSV

Clove





- Cloves are the dried flower buds of ***Syzygium aromaticum (Eugenia caryophyllus)*** *(Myrtaceae),*
- Clove oil contains **84–95% of phenols** (**eugenol** with about 3% of acetyleugenol), sesquiterpenes (α- and β-caryophyllenes).
- It is used as a flavouring agent, stimulant, aromatic and antiseptic.

Rosemary leaf



- whole leaf of *Rosmarinus officinalis L., family Labiatae.*
- contains 0.8-6% of esters and 8-20% of alcohols.
- The principal constituents are 1,8-cineole, borneol, camphor, bornyl acetate and monoterpene hydrocarbons, principally α-pinene and camphene.

Thyme

- leaves and flowers separated from the dried aerial parts of *T. vulgaris L. and T. zygis L., family Labiatae,*
- β-myrcene (1–3%), γ-terpinene (5–10%), *p-cymene (15–28%), linalol (4.0–6.5%), terpinen-4-ol* 0.2–2.5%), thymol (36–55%) and carvacrol (1.0 and 4.0%).

Cinnamon



- cinnamon is the dried bark of the shoots grown on cut stock of *Cinnamomum zeylanicum.*
- Oil of cinnamon contains about 60-75% w/w of *trans-cinnamic aldehyde,*
- Cinnamon contains volatile oil (*BP/EP not less than* 1.2%), phlobatannins
- Cinnamon is used as a flavouring agent and mild astringent

Summary

- Introduction
- Physical properties
- Methods of preparation
- Biosynthesis
- Chemistry
- Uses
- Distribution and occurrence

RESINS AND RESIN COMBINATIONS

By Million



INTRODUCTION

- The term 'resin' is applied to **more or less solid, amorphous products** of complex chemical nature.
- They are **insoluble in water** and **usually insoluble in petroleum ether** but dissolve more or less completely in alcohol, chloroform and ether.
- **Solid or semisolid plant** exudates produced normally (physiologically) or pathologically.
- On evaporation of resin solution **varnish like film** is deposited.



OCCURRENCE

- These are the amorphous mixtures of essential oils, **oxygenated products of terpenes** and carboxylic acids **found as exudations** from the trunk of various trees.
- Resins are produced **normally during growth** or **secreted as a result of injury** to the plant.
- End products of metabolism and considered as the oxidation product of **terpenes.**

GENERAL PROPERTIES OF RESINS

- Produce **smoky flame** on burning. E.g. Frankincense (Itan)
- On heating **they soften and finally melt**.
- **Hard, transparent or translucent** brittle materials.
- Amorphous materials but rarely crystallisable in nature.
- Resins are **bad conductors of electricity.**
- Heavier than water (Specific gravity ranging from 0.9-1.25).
- Purified resins are amorphous and brittle

Cont'd…. Solubility

- Resins are practically
  - insoluble in highly polar solvents (water) and highly non-polar solvents (pet. ether) but
  - dissolve in organic solvents like alcohol, chloroform, and ether.
- On evaporation of resin solution varnish like film is deposited.

WHY PLANTS PRODUCE RESINS?

- To seal plant wounds
- To kill insects and fungi
- To eliminate excess metabolites
- Reduce water loss
- Protect from UV radiation

CLASSIFICATION

- Resin and resin combinations are classified based on:

1. Based on source
2. Based on their combination with other constituents
3. Based on chemical nature
4. Based on the presence of major constituents

BASED ON SOURCE

- Resins commonly used in pharmacy are derived from <u>natural sources</u> and almost are **<u>plant products</u>** except **Shellac**, which an insect (Lac- *Kerria lacca*) secretion.





BASED ON THEIR COMBINATION WITH OTHER CONSTITUENTS

- On the basis of their occurrence in combination with other compounds, resins are classified into:
  a) Oleo resin
  b) Gum resin
  c) Oleo gum resin
  d) Glyco resin(gluco resin)
  e) Balsam

Cont'd….

A) Oleo resin

- When resins occur in combination with **volatile oils**, the mixture is called **oleoresin.**
- Volatile oil is removed by **steam distillation**
- E.g. Turpentine, **ginger**, capsicum, copaiba etc

B) Gum resin

- When resins occur in combination with gums
- Dissolving the associated gums in water can purify these resins
- E.g. Asafoetida, gamboge,

Cont'd….

C) OLEO-GUM RESIN

- Resins associated with both gums and volatile oils
- Volatile oil is removed by steam distillation while dissolving in water separates gum
- E.g. **Myrrh** and ipomoea

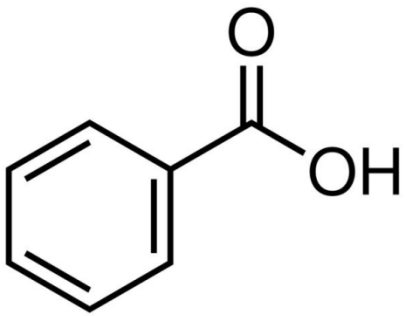
D) GLYCORESINS (GLUCORESIN)

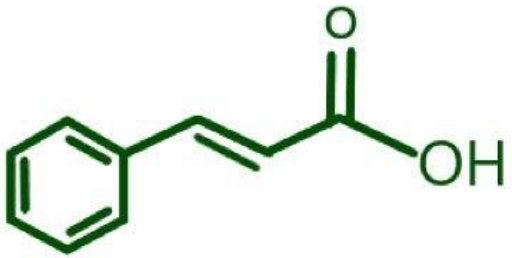
- Resins found in combination with glycosides
- On hydrolysis produce sugars and complex resins as aglycones.
- E.g. Resins which occur in ipomoea, jalap, and **podophyllum**

Cont'd….

E) BALSAMS

- Resinous substances containing high proportions of **balsamic acid (benzoic acid, cinnamic acid etc)** either in free or in combination or their esters.
- E.g. Tolu balsam contains 35 to 50% of balsamic acid, chiefly benzoic and cinnamic acid
- Benzoin, Peru balsam, and storax are examples of balsams.





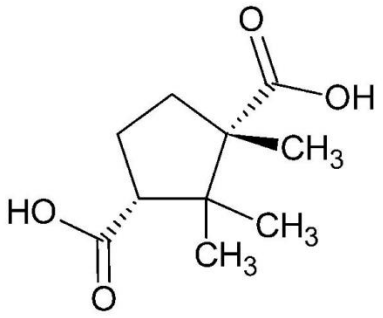
BASED ON CHEMICAL NATURE

- Chemically resins are amorphous products with a complex chemical nature.
- Are complex mixtures of the following compounds

1. Resin acids
2. Resin alcohols
3. Resenes

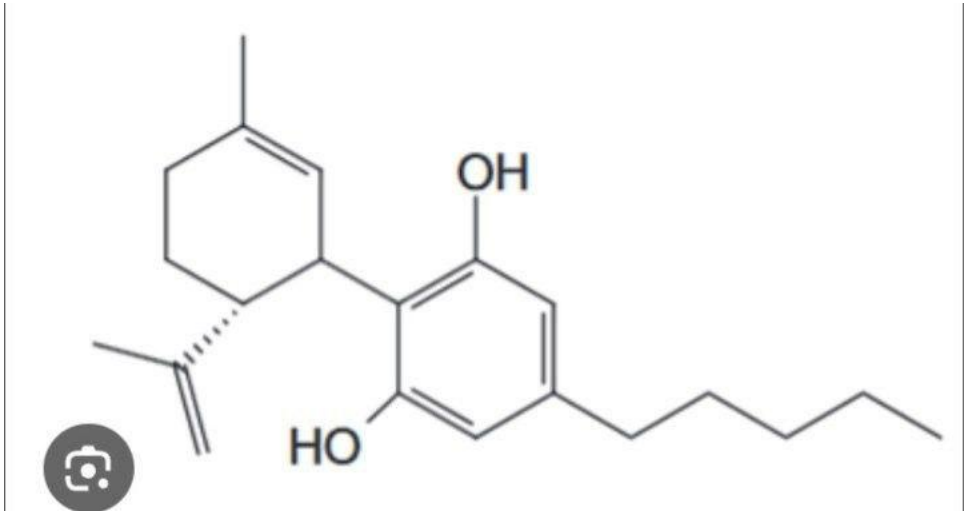
1. RESIN ACIDS

- Complex alcohols of high molecular weights
- Present in free state or as ester of simple aromatic acids like benzoic acid, cinnamic acid, and salicylic acid
- Commiphoric acid- Myrrh



2. Resinotannols (resin phenols)

- Form blue color with **ferric chloride**
- Have tannin like properties
- Example: Aloeresinotannol from aloe



2. RESIN ALCOHOLS OR RESINOL

- High molecular weight non phenolic alcohols
- Do not give tannin reaction with iron salts
- Example: Benzoresinol from benzoin

3. Resenes

- Complex neutral substances
- Do not respond to any chemical reactions
- Insoluble in alkalis and acids
- Do not form salts or esters
- Resist hydrolysis by alkalis
- Example: Alban and fluvial from **Gutta percha**

PREPARATION OF RESINS

- *Two general classes of resinous substances are recognized and this classification is based on the method used in preparing them:*

1. **Natural resins**, **occur as exudates from plants**, produced **normally or as result of pathogenic conditions**, as for example by artificial punctures e.g. mastic; or deep cuts in the wood of the plant e.g. turpentine, or by hammering and scorching, e.g. balsam of Peru.

Prepared resins

- are obtained by different methods.
- The drug containing ***resins*** is powdered and extracted with alcohol till exhaustion.
- The Concentrated alcoholic extract is **either evaporated**, or **poured into water** and the precipitated resin is collected, washed and carefully dried.
- In the preparation of ***oleoresins***; ether or acetone having lower boiling point are used. The volatile oil portion is removed through distillation.
- When the resin occurs associated with gum (***gum-resins***), the resin is extracted with alcohol leaving the gum insoluble.

CHEMICAL TESTS



IDENTIFICATION TEST	REAGENTS USED	POSITIVE RESULT	COMPOUNDS POSITIVE FOR THE TEST
HCl TEST	HCl	Formation of pink color	Presence of resins
FeCl3 TEST	Fecl3	Greenish blue color	Presence of resins
Combined Umbelliferone Test (Specific test for Asafoetida)	HCl + conc. NH4OH Nitric acid Sulfuric acid Washed with water	→Blue fluorescence →Green color →Red color →Violet	Umbelliferone
Copper Acetate test	Petroleum ether	Emerald green	Abietic acid

MEDICINAL USES OF RESINS

- Resin containing drugs posses many pharmacological activities
- Purgative - Podophyllum
- Cathartic - Gamboge, ipomoea
- Sedative - Cannabis
- Counterirritant - Capsicum
- Expectorant - Storax, benzoin

- Externally, resins are used as **mild antiseptics** in the form of cerates, ointments and plasters

Drugs containing resins

1. Podopyllum
2. Cannabis

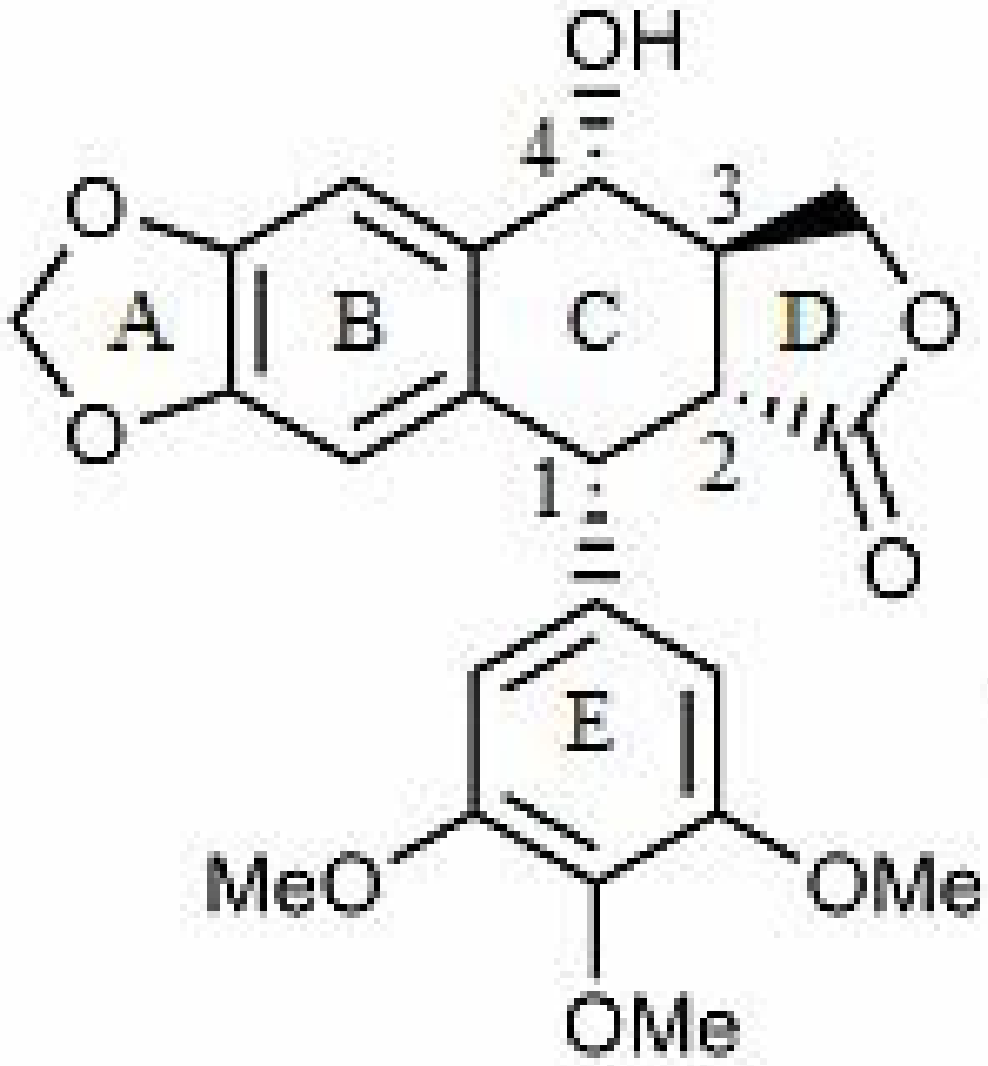
PODOPHYLLUM

- **Synonyms**: *Podophyllum radix*, Himalayan May-apple.
- **Biological source**: **Indian Podophyllum** consists of dried rhizomes and roots of **Podophyllum hexandrumor** P. emodi. **American podophyllum consists of dries rhizomes and roots of P. peltatum**.
- **Family**: Berberidaceae.
- **Geographical source**: *Podophyllum hexandrum* is found in India, China, and the Himalayas and hence known as Indian podophyllum, while *P. peltatum* (May apple or American mandrake) comes from North America and is known as American podophyllum.
- Plants are collected from the wild.

CHEMICAL CONSTITUENTS

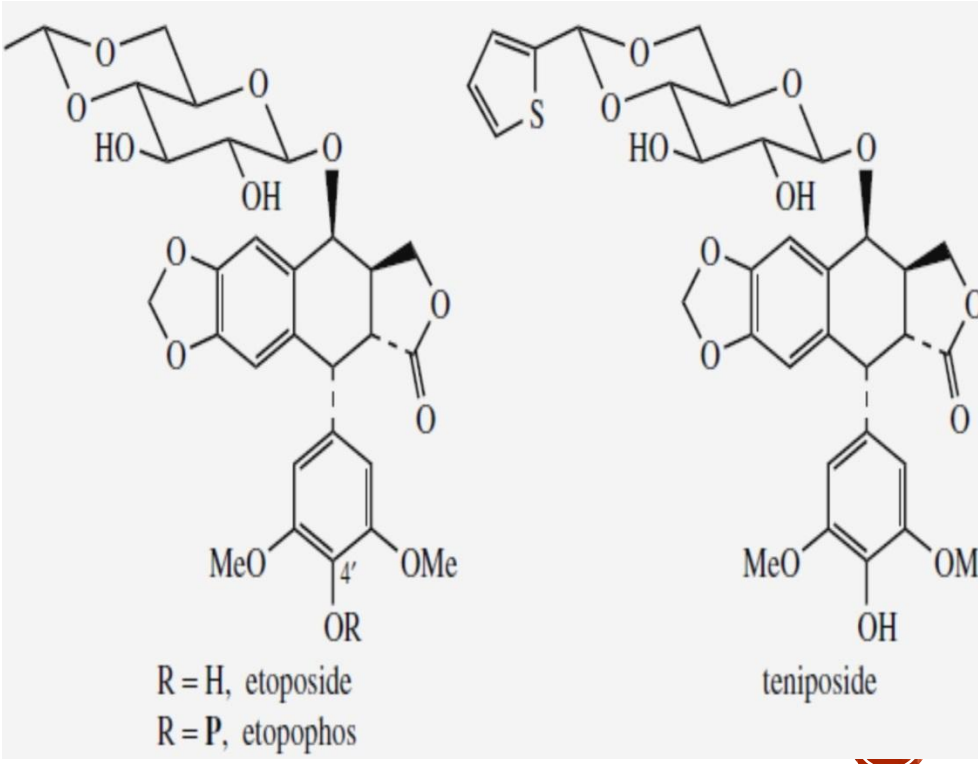
- The *roots contain* ***cytotoxic lignans*** and their glucosides, *P. hexandrum* containing about 5%, and *P. peltatum* about 1%.
- Indian podophyllum yields about 6–12% of resin containing **50–60% lignans, and American podophyllum 2–8% of resin containing 14–18%** ***lignans.***
- *The lignan constituents of the two roots are the same, but the proportions are markedly different*.

PODOPHYLLOTOXIN



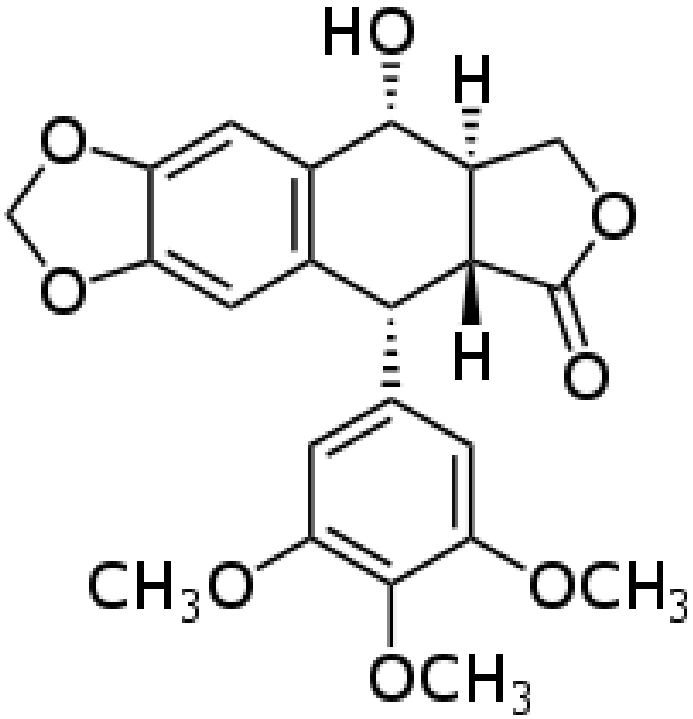
- The semi-synthetic derivatives **etoposide** and **teniposide** are manufactured from natural podophyllotoxin, and
  - Are excellent antitumour agents.

- Etoposide is a very effective anticancer agent, and is used in the treatment of **small cell lung cancer, testicular cancer and lymphomas**, usually in combination therapies with other anticancer drugs.



CHEMICAL CONSTITUENTS:

- **7-15% Resin known as podophyllin.**
- Podophyllin: Podophyllotoxin (present prominently in Indian variety); αand β-peltatins (present in American variety only).
- Podophyllin is a lignan chemically i.e. a substance whose chemical structure is based on 2,3-dibenzyl butane.
- Flavonoid: Astragalin.
- Others: Quercetin, kaempferol, essential oils.
- Etoposide and teniposide are semi-synthetic derivatives.



Podophyllotoxin

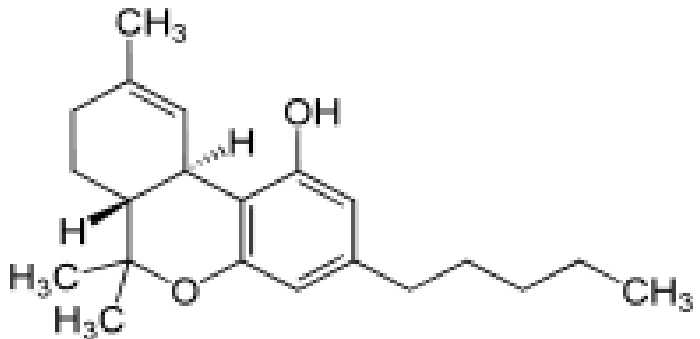
CANNABIS

- **Synonyms**: Indian hemp, Ganja, Marihuana
- **Biological source**: Cannabis consists of dried flowering tops of the cultivated female plants of ***Cannabis sativa*** Linn.
- **Family**: cannabinaceae.
- **Geographical source**:
  - It is indigenous to India, Commercially produced in Mexico and Africa.



Chemical constituents

- **15 to 20% of resin.**
- The major active principle of resin is **1-3-4 trans tetra-hydrocannabinol.**
- The resin also contains **cannabinol, cannabidiol, cannabidiolic acid, cannabichromene and cannabigerol**.
- It also contains trigonelline and choline, essential oils/volatile oils(pinene, myrcene, ocimene).



tetrahydrocannabinol

Uses

- It is a **narcotic, sedative and analgesic**.
- It has psychotropic properties due to tetrahydrocannabinol.
- It causes intoxication, euphoria and later mental disturbances.

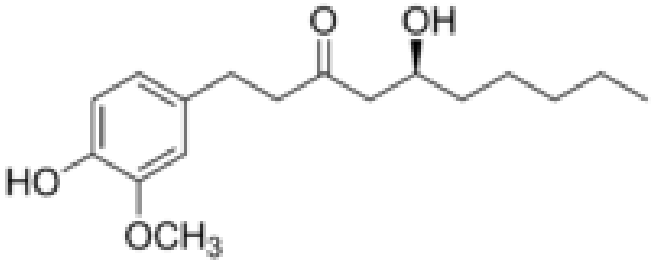
EXAMPLES OF DRUGS CONTAINING OLEORESIN

1. **Ginger**
2. **Capsicum**

Ginger

- **Synonyms**: zingiber.
- **Biological Source**: Ginger consists of whole or cut, dried scrapped or **unscrapped rhizomes of *Zingiber officinale*** Roscoe.
- **Family**: Zingiberaceae
  - It contains **0.8% of total gingerols on dried basis**.
- **Geographical Source**:
  - Native to South-east Asia
  - Cultivated in Caribbean islands, Africa, Australia, Mauritius, Jamaica, Taiwan and India.

Chemical constituents



Gingerol

- **5 to 8% resinous mass.**
- 1-2% volatile oil: Responsible for aromatic odor of drug.
  - **Gingerol-Responsible for pungency of the drug**.
  - Sesquiterpene hydrocarbon-α-zingiberol, α-farnesene, α-bisabolene, α-sesquiphellandrene.
  - Less pungent components: Gingerone; Shogoal.
  - Shogaol: Not present in fresh rhizome; Formed by dehydration of gingerol.

Shogaol

Zingerone

Uses

- Stomachic, an aromatic, carminative & a stimulant.
- Flavoring agent.
- Effective in motion sickness.
- Positive inotropic effect.
- Spasmolytic.
- Effective in sore throat, cold, cough & in asthma.

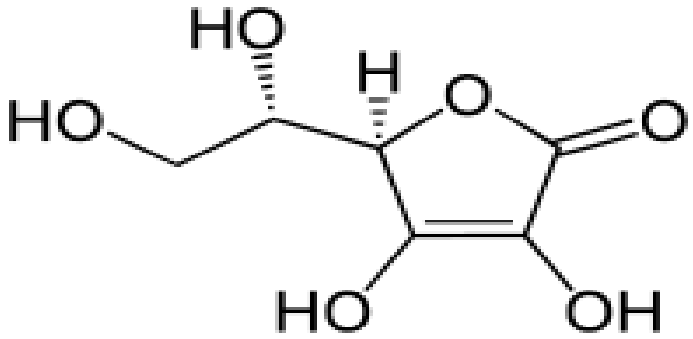
Capsicum

- **Synonyms**: Chillies, Cayenne pepper.
- **Biological Source**:
  - Capsicum consists of **dried ripe fruits of *Capsicum annum*** Linn. var minimum.
  - Capsicum or cayenne pepper is the dried ripe fruit of Capsicum frutenscens, known in commerce African chilies.
  - It contains not less than 12% of non-volatile ether soluble extractives.
- **Family**: Solanaceae.
- **Geographical Source**: All tropical countries-E. Africa, W. Africa and India.

Chemical Constituents

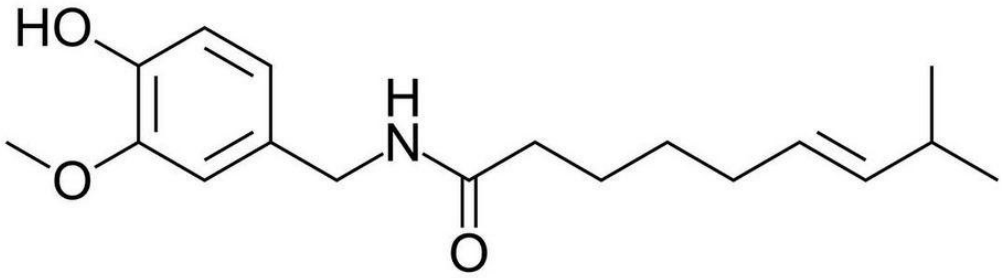
- **0.5 to 0.9% colourless, crystalline and pungent principle known as Capsaicin(amide).**
- 0.1 to 0.5% Ascorbic acid.
- **Capsainthin** and Carotene-pigments responsible for red color. **It also contains volatile oils (1.5%),** carotenoids.





Ascorbic acid





Capsaicin

$C_{18}H_{27}NO_3$

Uses

- Used in culinary preparations.
- Carminative, appetizer and a stomachic.
- Externally, used as counter-irritant: in rheumatism, inflammation, lumbago, neuralgia.
- Used in spices.



EXAMPLES OF DRUGS CONTAINING OLEO GUM RESIN

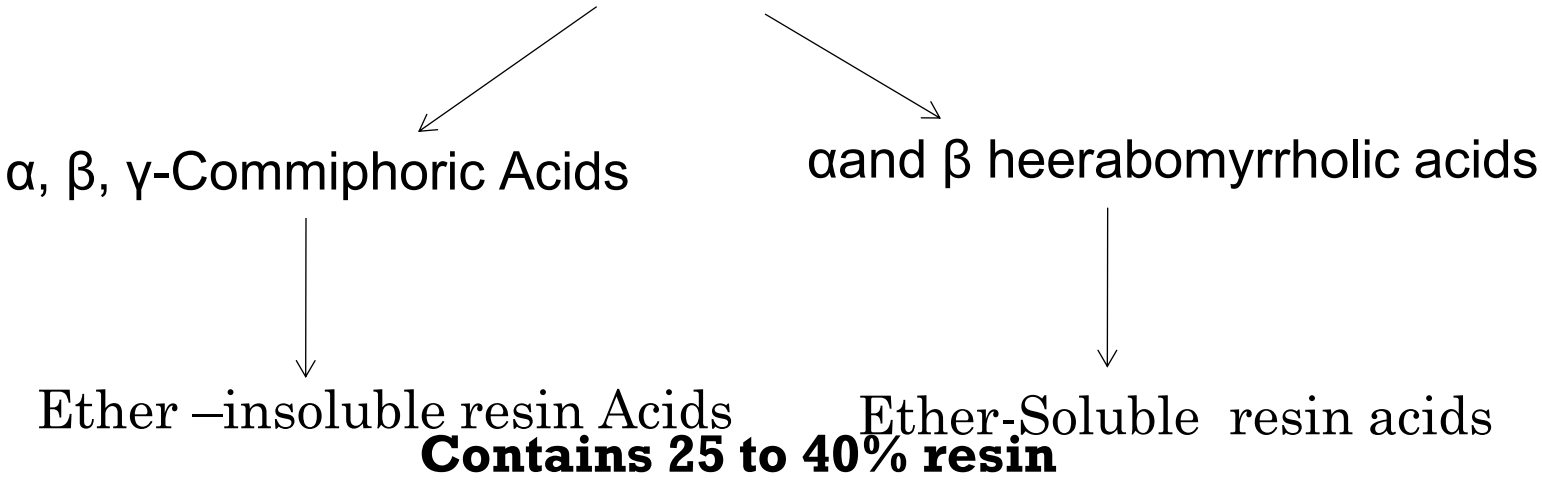
1. **Myrrh**



Myrrh

- **Synonyms**: Myrrha, Hirabol, Bol.
- **Biological Source**: Myyrh is an oleo-gum-resin obtained from ***Commiphora molmol*** Engler and from other Commiphora species.
- **Family**: **Burseraceae**.
- **Geographical Source**: North-east Africa, Southern Arabia.
- Odour: Aromatic and agreeable.

Myrrh: Chemical Constituents:



Contains 25 to 40% resin

- 10% yellowish thick volatile oil-terpenes, cuminic aldehyde, **eugenol.**
- **60% gum**. Gum is associated with oxidase enzyme.
- **A bitter principle.**

BALSAMS

- Balsams are resinous mixtures that contain large **proportions of cinnamic acid, benzoic acid or both or esters of these acids**.
- The term "balsam" is often wrongly applied to oleoresins and should be reserved for such substances as *balsam of Peru*, *balsam of Tolu* and *storax*, which contain a high proportion of aromatic balsamic acids.

BALSAMS

- Balsams are naturally occurring resinous mixtures, which contain high proportion of aromatic balsamic acids i.e. BENZOIC ACID and CINNAMIC ACID and their esters.
- Balsams containing free acids are partially soluble in hot water.
- Some important balsam containing crude drugs are:-

- Balsam of Peru,
- Balsam of Tolu,
- Benzoin and Storax.

TOLU BALSAM

- **Synonym:** Balsamum tolutanum
- Tolu balsam is obtained from the incisions made on the tree trunk of *Myroxylon balsamum*.
- It has pleasant odor and aromatic balsamic taste.
- It is soluble in alcohol; it gives green color with $FeCl_3$ solution and it produces the odor of benzaldehyde due to the oxidation of cinnamic acid.
- It contains about 80% of resin alcohols in combination with the balsamic acids.
- It contains free balsamic acids, benzyl benzoate, benzyl cinnamate, ferulic acid, vanillin and eugenol.
- Balsam of Tolu is used as an important ingredient of cough preparations.

PERU BALSAM

- **Synonym:** Balsamum peruviamum
- Peru balsam is derived from the injuries in the trunk of *Myroxylon pererae*.
- It has pleasant aromatic vanilla like odor and acrid bitter taste.
- Peru balsam contains esters of the balsamic acids (benzyl benzoate, benzyl cinnamate, and cinnamyl cinnamate) and peruresinotannol.
- It is used as an antiseptic and parasiticide in certain skin diseases such as scabies.
- It is also used for the treatment of haemorrhoids due to its astringent properties and as an ingredient in paraffin gauge dressings.

CRUDE DRUGS CONTAINING RESINS (FOCUS ON THE BOLD ONES

Name of Crude drug	Biological Source	Chemical Constituents	Uses
Podophyllum (Indian apple)	***Podophyllum emodi,* Family: Berberidaceae**	**Podophyllotoxin, Picropodophyllin Flavones starch**	**Anticaner agent**
Asafoetida (Devil's dong)	*Ferula foetida*, Family: Umbelliferae,	Ferolic acid, Asaresinol Pinene	Stimulant,Carminative
Ginger, (Zingiber)	***Zingiber afficinale,* Family: Zingiberaceae**	**Zingiberine, Curcumene, Phenyl propanoids**	**Stimulant, Aromatic, Carminative**
Turmeric (Indian saffron)	*Curuma longa* Family: Zingiberaceae	Curcumine -I, II, III, Sugar, Camphene	Condiment, Colouring agent
Capsicum (Red pepper)	***Capsicum annum* Family: solanacea**	**Capsaicin, Casanthin, Carotin Protein, Ascorbic acid**	**Stomachic, Stimulant**
Benzoin (Loban)	*Styrax benzoin* Family: Styraceae	Siaresinolic acid, Sumaresinolic acid Vanillin, Cinnamic acid	Diuretic, Expectorant, Carminative

Name of Crude drug	**Biological Source**	**Chemical Constituents**	**Uses**
Colophony (Rosina)	*Pinus elliotti*, Family: Pinaceae	Abietic acid Pimaric acid, Sapinic acid	Stimulant, Diuretic
Shellac (Lac)	Lcifer lacca Family: Lacciferidae Order: Hemiptera	Resin, shelloic acid aleuritic acid, kerrolic acid, butolic acid	Sustained release medicaments
Cannabis (Indian hemp)	***Cannabis Sativa,* Family: *Cannabinaceae***	**Tetrahydro-cannabinaol, Cannabinaol Trigonelline Choline**	**Antitussive, Antianxiety,**
Guggul	*Commiphora weightii* Family: Burseraceae	Gum resin, steroids , guggulostrones	Antiinflammatory, Hypolipidemic